Experiment No :- 1.

- Aim :-

To design & implement a relational database scheme using my SQL, postgre SQL for a student Enrollment system. ensuring data integrity through primary & foreign key constraints.

Theory :-

- Data base

A database is an organized collection of related data that allows efficient storage, retrieval & management of information.

- Relational Database

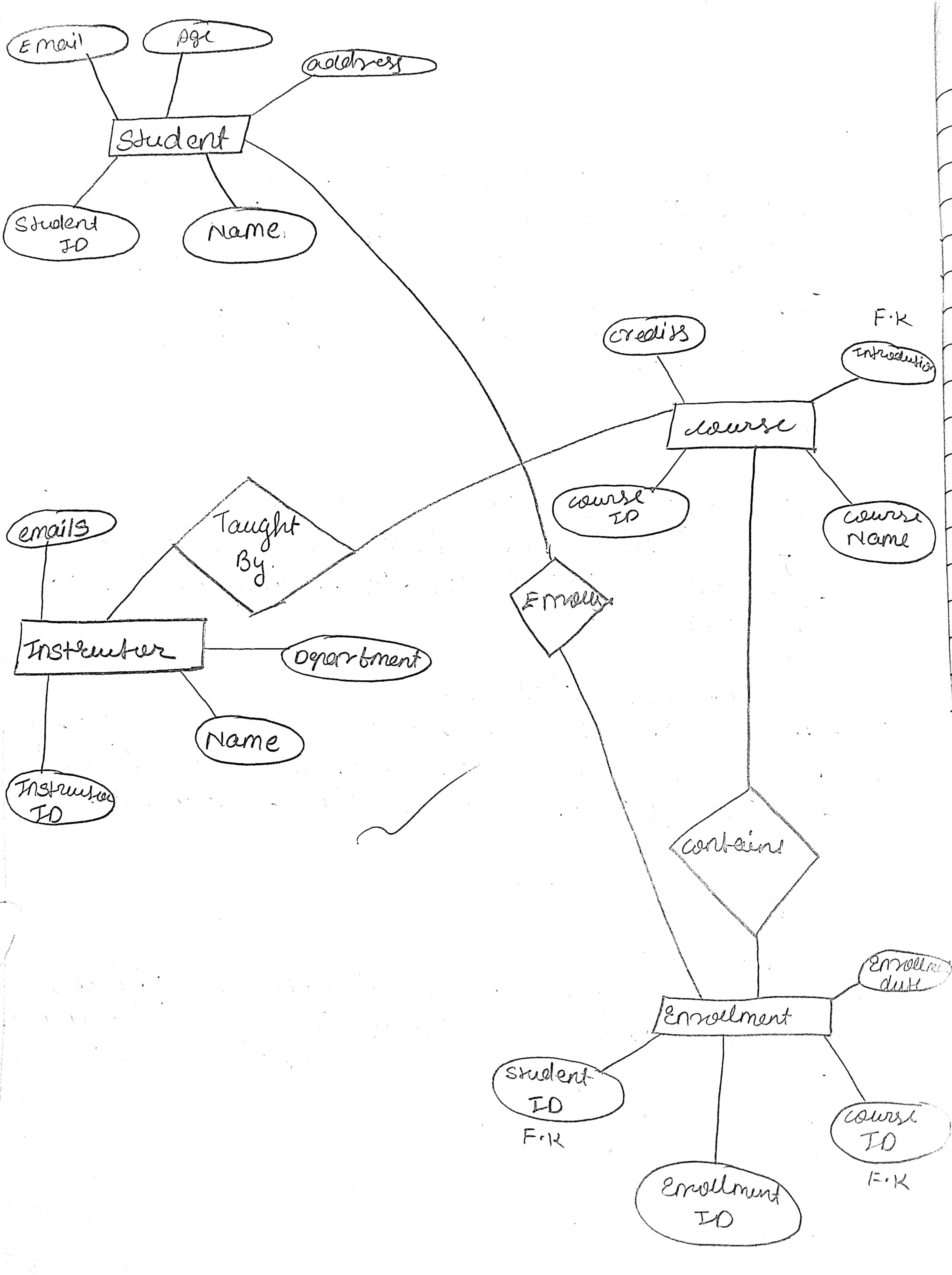
A relational database stores data in the form of tables (relations) consisting of rows & columns.

Key components :-

1] Entities :-

Entities represent real-world objects whose data is stored in the database. In this system the entities are :

- Student
- course
- Instructor
- Enrollment.



Email
age
address
Student
Student ID
Name
credits
F.K
Introduction
course
course ID
course Name
emails
Taught By
Enrolls
Instructor
Department
Name
Instructor ID
contains
Enrollment date
Enrollment
Student ID
F.K
course ID
F.K
Enrollment ID

2] Relationships :-
Relationships describes how entities are connected.
- A student can enroll in multiple courses
- A course can have many students
- This forms a many to many relationship between students & course
- The enrollment tables is used to resolve this relationship.
- An Instructor teaches one or more course.

3] Keys :-
- Primary key :-
It uniquely identifies each record in a table (eg Student ID)
- Foreign key :- It creates relationship between tables.
eg (Instructor ID → Foreign key in course table).

4] Attributes :-
It is defined as properties describing an entity
eg :- Attributes of student are :- Student ID, Name email age address.

* Entity - Relationship (E-R diagram)
The ER diagram acts as the blueprint. It illustrates how the entities interact. In this system :

- An instructor has one to many relationship with causes (one instructor can teach many courses).
- Student & course have a many to many relationship which is resolved a junction table or link table named enrollment.

- Conclusion :-

The database schema for the student enrollment system was successfully designed & implemented by using primary keys, we ensured record uniqueness & by using foreign keys, we established logical relationships between entities, effectively mapping the E-R diagram into a functional SQL database.

14/3/26

Practical no 1

Program:

```
CREATE DATABASE StudentDB;
USE StudentDB;
CREATE TABLE Student(
StudentID INT PRIMARY KEY,
Name VARCHAR(50),
Email VARCHAR(50),
Age INT,
Address VARCHAR(100)
);
CREATE TABLE Instructor(
InstructorID INT PRIMARY KEY,
Name VARCHAR(50),
Email VARCHAR(50),
Department VARCHAR(50)
);
CREATE TABLE Course(
CourseID INT PRIMARY KEY,
CourseName VARCHAR(50),
Credits INT,
InstructorID INT,
FOREIGN KEY (InstructorID) REFERENCES Instructor(InstructorID)
);
CREATE TABLE Enrollment(
EnrollmentID INT PRIMARY KEY,
```

```
StudentID INT,
CourseID INT,
EnrollmentDate DATE,
FOREIGN KEY (StudentID) REFERENCES Student(StudentID),
FOREIGN KEY (CourseID) REFERENCES Course(CourseID)
);

INSERT INTO Student VALUES
(1,'Amit','amit@gmail.com',20,'Pune'),
(2,'Neha','neha@gmail.com',21,'Mumbai');
INSERT INTO Instructor VALUES
(101,'Dr. Rao','rao@gmail.com','Computer'),
(102,'Dr. Sharma','sharma@gmail.com','IT');
INSERT INTO Course VALUES
(201,'DBMS',4,101),
(202,'Operating Systems',3,102);
INSERT INTO Enrollment VALUES
(1,1,201,'2025-02-01'),
(2,2,202,'2025-02-02');

SELECT * FROM Student;
SELECT * FROM Instructor;
SELECT * FROM Course;
SELECT * FROM Enrollment;
```

Output:

Result Grid | Filter Rows: | Edit: | Export/Import

StudentID	Name	Email	Age	Address
1	Amit	amit@gmail.com	20	Pune
2	Neha	neha@gmail.com	21	Mumbai
NULL	NULL	NULL	NULL	NULL

Result Grid | Filter Rows: | Edit: | Export/Import

InstructorID	Name	Email	Department
101	Dr. Rao	rao@gmail.com	Computer
102	Dr. Sharma	sharma@gmail.com	IT
NULL	NULL	NULL	NULL

Result Grid | Filter Rows: | Edit: | Export/Import

CourseID	CourseName	Credits	InstructorID
201	DBMS	4	101
202	Operating Systems	3	102
NULL	NULL	NULL	NULL

Result Grid | Filter Rows: | Edit: | Export/Import

EnrollmentID	StudentID	CourseID	EnrollmentDate
1	1	201	2025-02-01
2	2	202	2025-02-02
NULL	NULL	NULL	NULL